

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

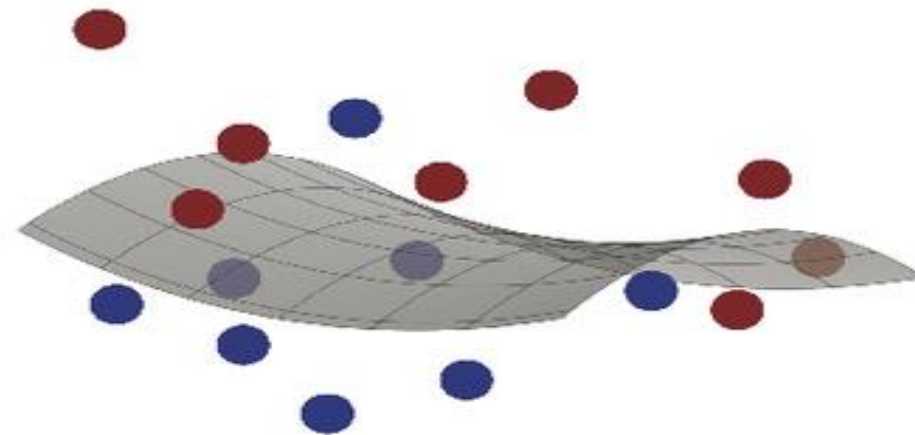
Reinforcement Learning

Texas A&M University

Additional Materials

- Reinforcement Learning

Foundations of
Machine Learning second edition



Mehryar Mohri,
Afshin Rostamizadeh,
and Amrith Talwalkar

- Source:

Additional Source:

Dr. Guni Sharon's
reinforcement
Learning course

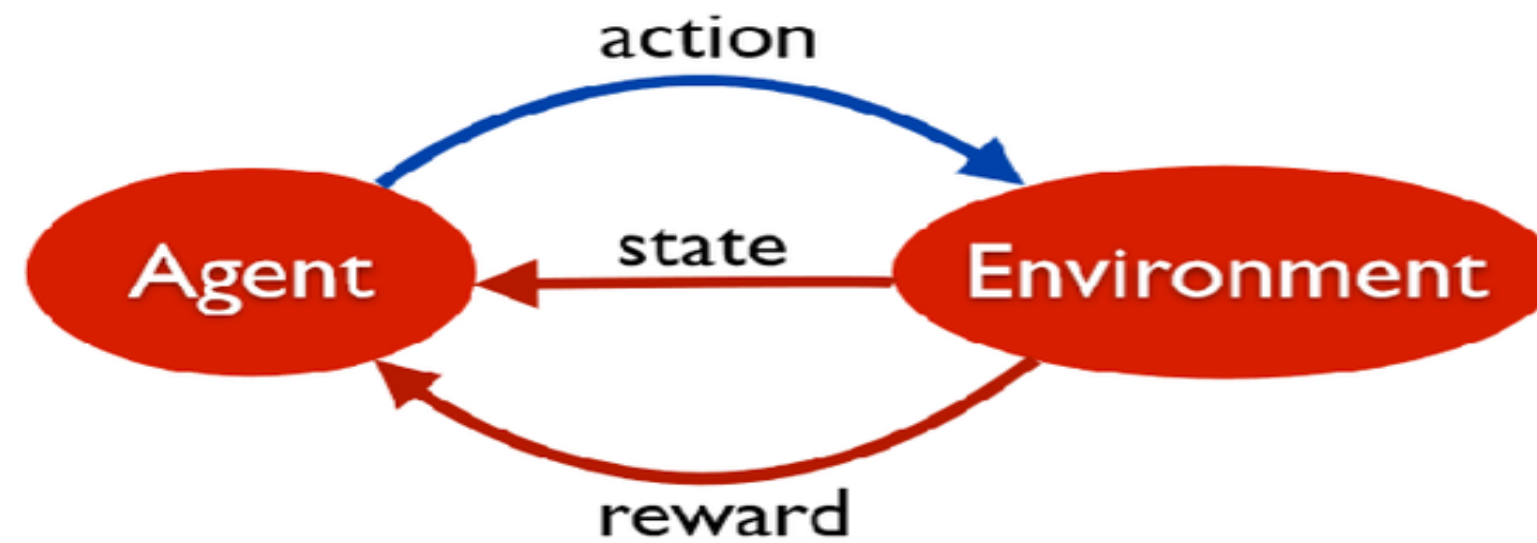
Topics

- What is reinforcement learning?
- An example application and Markov Decision Process
- Value Iteration
- Q Learning (and Policy Iteration)

Reinforcement Learning

- We have an **Actions** and **Environment** that do not passively collected labeled data
- The learner, called an **Agent**, gets two kinds of information to learn from: The current state of the environment, and a real-valued reward
- The objective of the learning problem is for the agent to maximize its reward
- It does this through finding the best course of actions - called a **policy**

Reinforcement Learning



- Exploration - search unknown states and actions to gain reward information
- Exploitation - search known states to optimize reward

Reinforcement Learning vs. Supervised Learning

- No fixed distribution that instances are drawn from
- Environment may not be fixed!
- Training and testing phases are mixed.
- Planning Problem: When the environment model is known - objective is to maximize reward
- Learning Problem: Environment model is unknown
- We will explore both

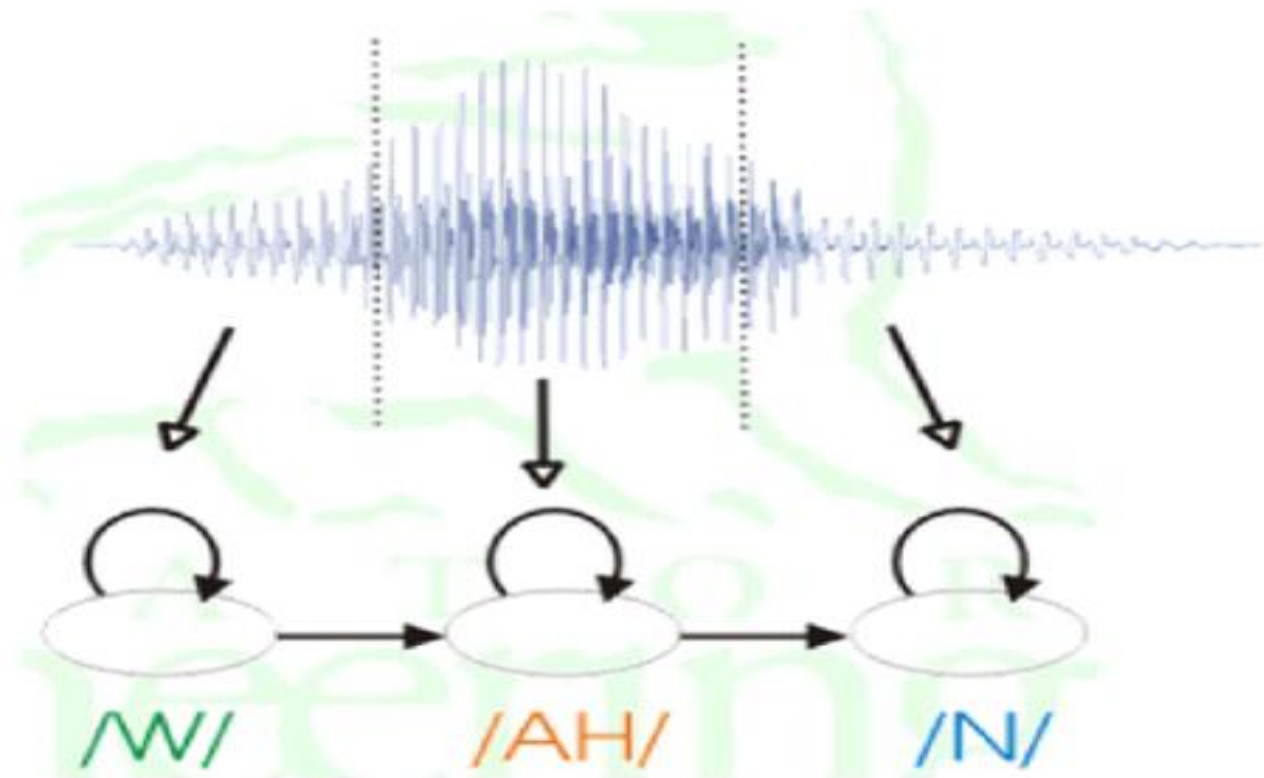
Applications

- Robot control e.g., Robocup Soccer Teams (Stone et al., 1999).
- Board games, e.g., TD-Gammon (Tesauro, 1995).
- Elevator scheduling (Crites and Barto, 1996).
- Ads placement.
- Telecommunications.
- Inventory management.
- Dynamic radio channel assignment.

Motivation

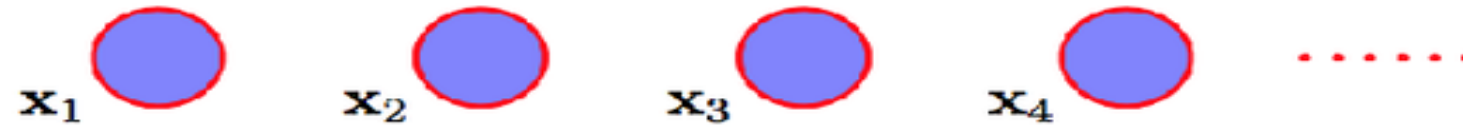
Why to model time-series?

- Many phenomena depict inherent dependencies between successive time points
- Examples
 - speech, DNA sequencing, industrial processes, human behavior
- Data samples are no longer considered iid

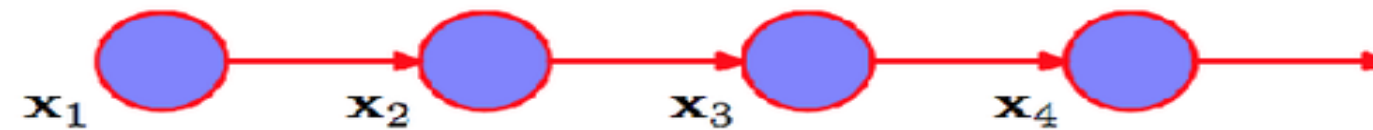


Discrete (Observable) Markov Model

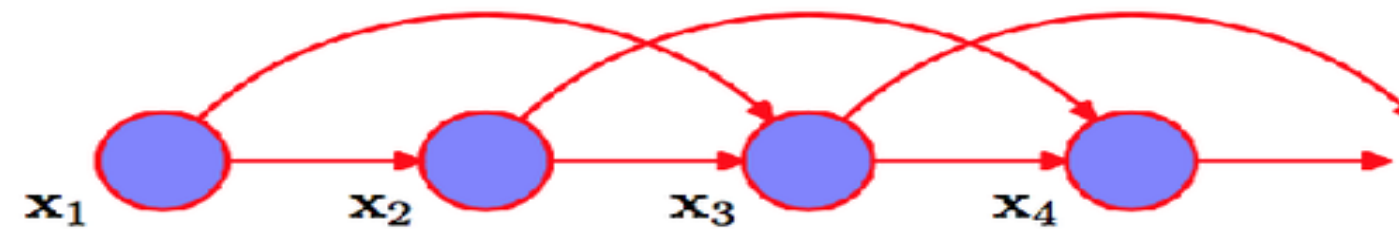
The simplest approach to modelling a sequence of observations is to treat them as independent, corresponding to a graph without links.



A first-order Markov chain of observations $\{x_n\}$ in which the distribution $p(x_n|x_{n-1})$ of a particular observation x_n is conditioned on the value of the previous observation x_{n-1} .



A second-order Markov chain, in which the conditional distribution of a particular observation x_n depends on the values of the two previous observations x_{n-1} and x_{n-2} .



For the next slides, $q_t = x_t$ is the i^{th} observable sample of the sequence $Q = \{q_1, q_2, \dots, q_T\}$

where $q_t \in \{S_1, \dots, S_N\}$

Discrete (Observable) Markov Model

- **N distinct states:** $\{S_1, S_2, \dots, S_N\}$
- **Observable sequence:** $Q = \{q_1, q_2, \dots, q_T\}$
- $q_t = S_i$: at time t the system is at state S_i
- General Markov model: future state depends on current & previous

$$P(q_{t+1} = S_j | q_t = S_i, q_{t-1} = S_k, \dots)$$

- 1st-order Markov model: future state depends only on current

$$P(q_{t+1} = S_j | q_t = S_i)$$

The future is independent of the past

Discrete (Observable) Markov Model

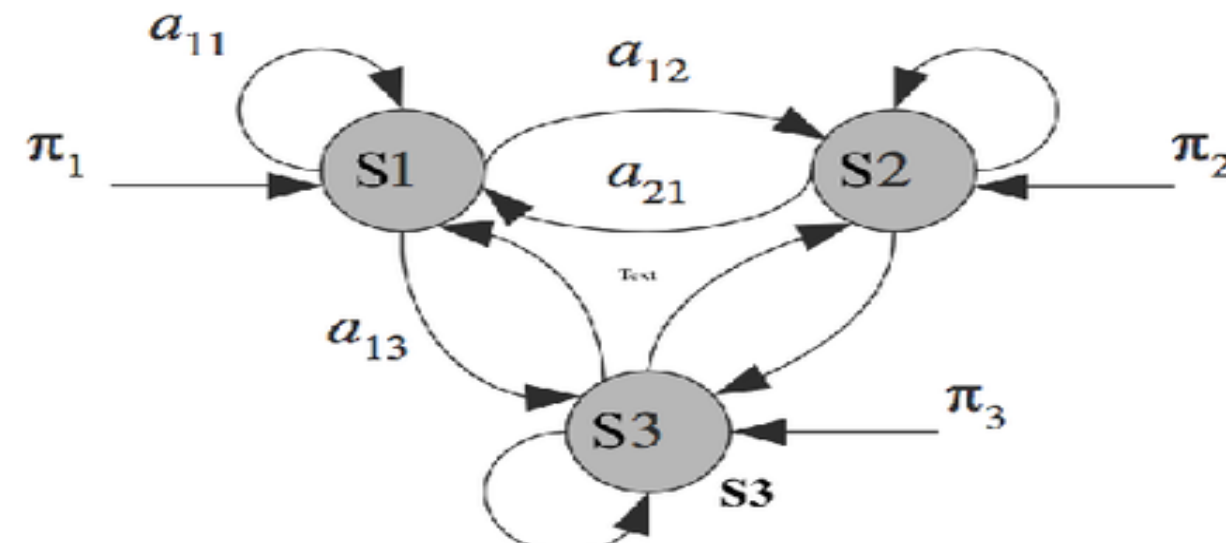
- Constant transition probability

$$\alpha_{ij} = P(q_{t+1} = S_j | q_t = S_i) , \quad \alpha_{ij} \geq 0 , \quad \sum_{j=1}^N \alpha_{ij} = 1$$

Going from S_i to S_j has the same probability no matter when it happens

- Initial probability

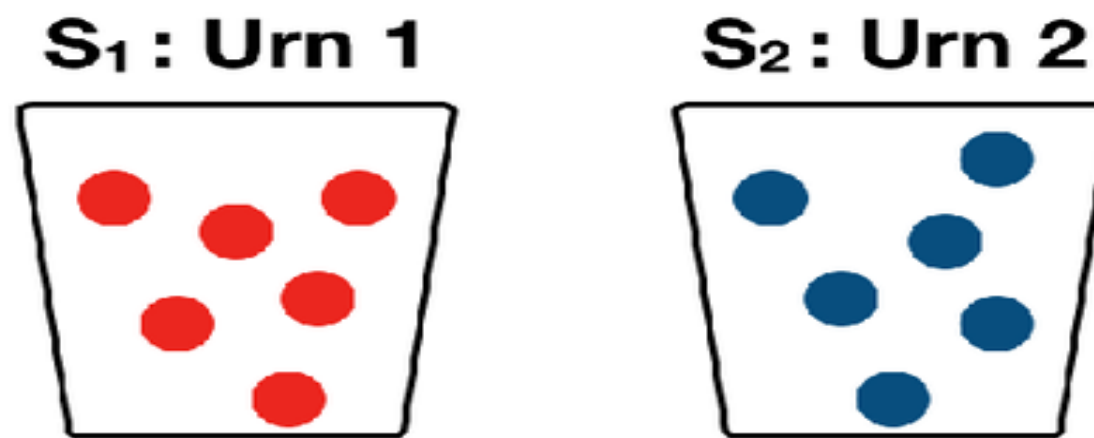
$$\pi_i = P(q_1 = S_i) , \quad \sum_{j=1}^N \pi_i = 1$$



Discrete (Observable) Markov Model

Observable Markov model

- The states are *observable*
 - We know q_t at any time t
- Transition matrix $\mathbf{A} = [\alpha_{ij}]$
- Initial probability vector $\boldsymbol{\pi} = [\pi_1, \dots, \pi_N]$



A horizontal sequence of eight dots: red, red, blue, red, blue, blue, red, red.

$$Q = \{q_1, q_2, \dots, q_8\}$$

Discrete (Observable) Markov Model

Observable Markov model: Two Basic Problems

- 1 Given a model $\lambda = \{\mathbf{A}, \pi\}$, we would like to evaluate the probability of a given observation sequence $Q = \{q_1, \dots, q_T\}$: $P(\mathbf{Q}|\mathbf{A}, \pi)$
- 2 Given a training set of observation sequences, $\mathcal{X} = \{Q^k\}_{k=1}^K$, we would like to learn the model that maximizes the probability of generating $\mathcal{X}$: $P(\mathcal{X}|\mathbf{A}, \pi)$

Discrete (Observable) Markov Model

Observable Markov model: Two Basic Problems

Problem 1

Given a model $\lambda = \{\mathbf{A}, \pi\}$, we would like to evaluate the probability of a given observation sequence $Q = \{q_1, \dots, q_T\}$: $P(\mathbf{Q}|\mathbf{A}, \pi)$

$$P(\mathbf{Q}|\mathbf{A}, \pi) = P(q_1) \prod_{t=2}^T P(q_t|q_{t-1}) = \pi_{q_1} \alpha_{q_1 q_2} \cdots \alpha_{q_{T-1} q_T}$$

Discrete (Observable) Markov Model

Observable Markov model: Two Basic Problems

Problem 2

Given a training set of observation sequences, $\mathcal{X} = \{Q^k\}_{k=1}^K$, we would like to learn the model that maximizes the probability of generating $\mathcal{X}$:

$P(\mathcal{X}|\mathbf{A}, \pi)$

$$\hat{\pi}_i = \frac{\text{\#sequences starting with } S_i}{\text{\#sequences}} = \frac{\sum_{k=1}^K \mathbb{I}(q_1^k = S_i)}{K}$$

$$\hat{\alpha}_{ij} = \frac{\text{\#transitions from } S_i \text{ to } S_j}{\text{\#transitions from } S_i} = \frac{\sum_{k=1}^K \sum_{t=1}^{T-1} \mathbb{I}(q_t^k = S_i \text{ and } q_{t+1}^k = S_j)}{\sum_{k=1}^K \sum_{t=1}^{T-1} \mathbb{I}(q_t^k = S_i)}$$

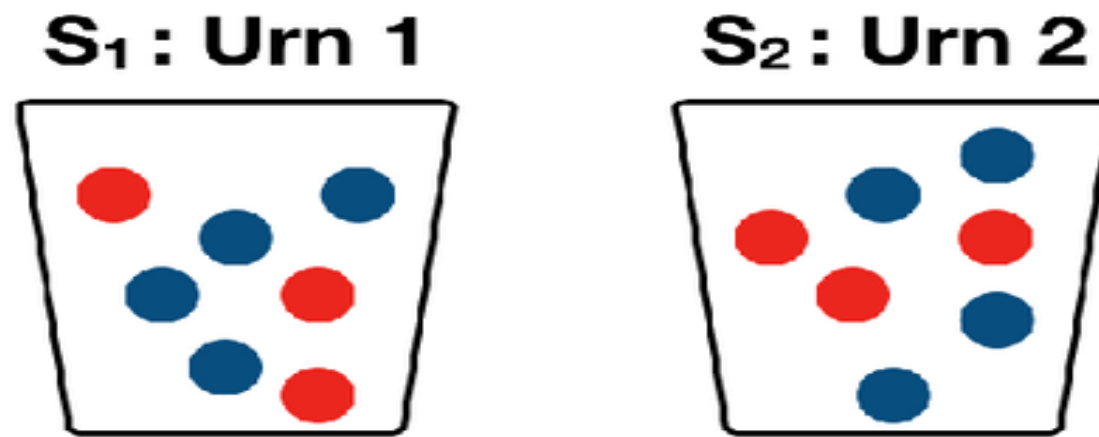
Hidden Markov Model

- The states are *not* observable
- But when we reach a state, an observation occurs with *emission probability*

$$b_j(m) = P(O_t = v_m | q_t = S_j)$$

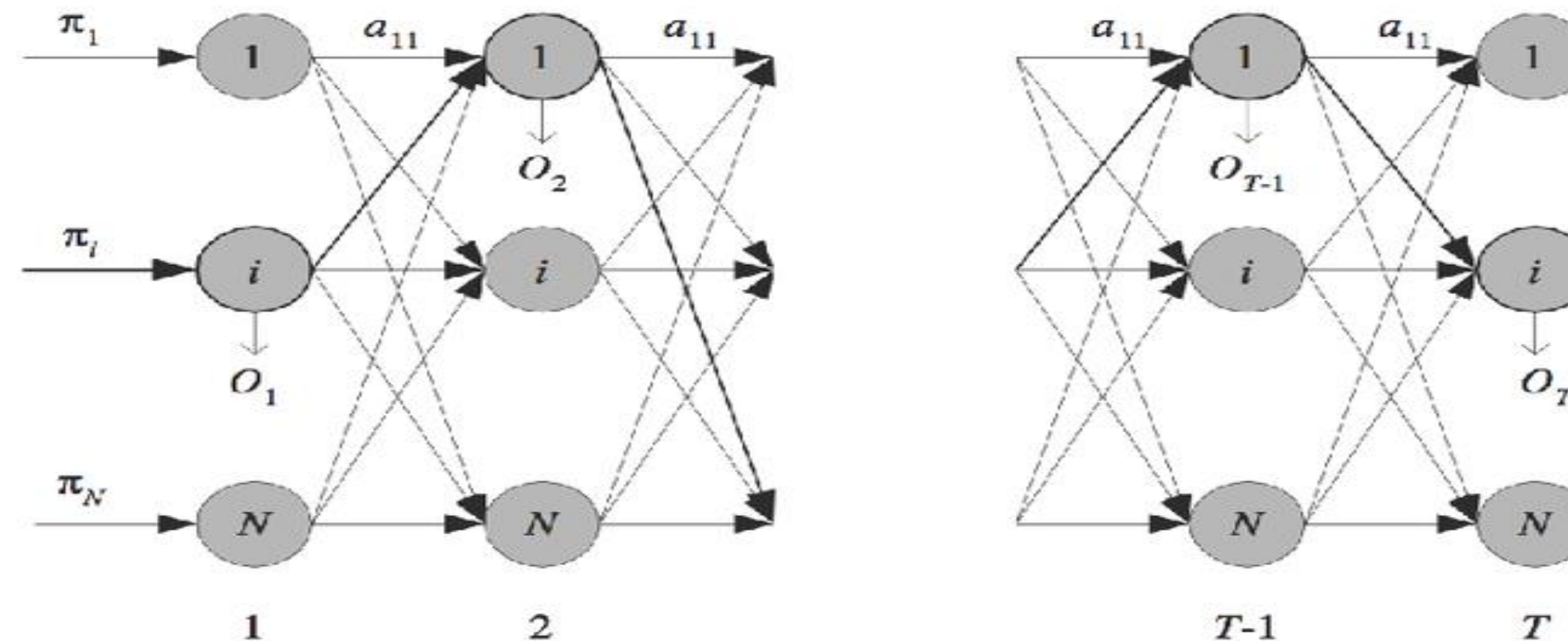
$b_j(m)$: is the probability of observing value v_m in state S_j

- Example: each urn contains balls of different colors
 - $b_j(m)$ is the probability of drawing a ball of color m from urn j



Hidden Markov Model

- Hidden states: $\{S_1, \dots, S_N\}$
- Observation symbols: $\{v_1, \dots, v_M\}$
- State transition probabilities: $\mathbf{A} = [a_{ij}]$, $a_{ij} = P(q_{t+1} = S_j | q_t = S_i)$
- Observation probs: $\mathbf{B} = [b_j(m)]$, $b_j(m) = P(O_t = v_m | q_t = S_j)$
- Initial state probabilities: $\boldsymbol{\pi} = [\pi_i]$, $\pi_i = P(q_1 = S_i)$



Hidden Markov Model

Hidden Markov model: Three Basic Problems

- 1 Given a model $\lambda = \{\alpha_{ij}, \pi_i, b_j(m)\}$, we would like to evaluate the probability of a given observation sequence $O = \{O_1, \dots, O_T\}$:
 $P(\mathbf{O}|\lambda)$
- 2 Given a model λ and observation sequence O , we would like to find out the state sequence $Q = \{q_1, \dots, q_T\}$, that generates O with the highest probability: $Q^* = \max_Q P(Q|O, \lambda)$
- 3 Given a training set of observation sequences, $\mathcal{X} = \{O^k\}_{k=1}^K$, we would like to learn the model that maximizes the probability of generating $\mathcal{X}$: $\lambda^* = P(\mathcal{X}|\lambda)$

Reinforcement Learning: Markov Decision Process

- Set of epochs $\{0, \dots, T\}$
- a set of states S , possibly infinite!
- an initial state $s_0 \in S$
- Actions A , also possibly infinite
- Transition Probability $P(s'|s, a)$ which is the distribution over destination states $s' = \delta(s, a)$
- Reward Probability $P(r'|s, a)$ which is the distribution over rewards returned $r' = r(s, a)$

Policy

- MDP wants to determine what action to take at each state - that is - the policy
- $\pi : S \rightarrow \Delta(A)$ where $\Delta(A)$ is the set of probability distributions of actions A .
- A policy is deterministic if, for any s , there exists a unique $a \in A$ such that $\pi(s)(a) = 1$
- Stationary Policy - choice of distribution of actions does not depend upon time.

Policy: Non-stationary

- Finite horizon ($T < \infty$): $\sum_{t=0}^T r(s_t, \pi(s_t))$
- Infinite horizon ($T = \infty$): $\sum_{t=0}^{+\infty} \gamma^t r(s_t, \pi(s_t))$ where $\gamma^t \in [0, 1)$ (discounts future rewards - earlier rewards are more important)

Policy: Value

- Value $V_{\pi}(s)$ is:
- Finite horizon: $V_{\pi}(s) = \mathbb{E}_{a_t \sim \pi(s_t)} \left[\sum_{t=0}^T r(s_t, a_t) | s_0 = s \right]$
- Infinite horizon: $V_{\pi}(s) = \mathbb{E}_{a_t \sim \pi(s_t)} \left[\sum_{t=0}^{+\infty} \gamma^t r(s_t, a_t) | s_0 = s \right]$
- These expectations are over the random selection of an action a_t according to a distribution $\pi(s_t)$
- We want to find a policy π that maximizes value over all states.
- A policy π^* is optimal if its value is maximal for every state s .
- For any MDP there exists a deterministic optimal policy.

State Action Value Function

- Q state-action value function associated to a policy π for all $(s, a) \in S \times A$
- $Q_{\pi}(s, a) = \mathbb{E}[r(s, a)] + \mathbb{E}_{a_t \sim \pi(s_t)}[\sum_{t=0}^{+\infty} \gamma^t r(s_t, a_t) | s_0 = s, a_0 = a]$
- We find that $\mathbb{E}_{a \sim \pi(s)}[Q_{\pi}(s, a)] = V_{\pi}(s)$
- This is by the Bellman Equations:
$$V_{\pi}(s) = \mathbb{E}[r(s, \pi(s))] + \gamma \sum_{s'} \mathbb{P}[s' | s, \pi(s)] V_{\pi}(s')$$
- What does all this mean? That an optimal policy exists and can be determined. (Existence and Uniqueness)